1)WRITE A JAVA PROGRAM TO DEMONSTRATE THE USE OF TRY, CATCH,AND FINALLY BLOCKS FOR HANDLING MULTIPLE BUILT-IN EXCEPTIONS

PROGRAM CODING:

```
import java.util.Scanner;


public class Main {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    try {
      System.out.print("Enter numerator: ");
      int a = sc.nextInt();


      System.out.print("Enter denominator: ");
      int b = sc.nextInt();


      int result = a / b;
      System.out.println("Result = " + result);


      int[] arr = {10, 20, 30};


      System.out.print("Enter array index: ");
      int index = sc.nextInt();


      System.out.println("Array value = " + arr[index]);


    } catch (ArithmeticException e) {
      System.out.println("Error: Cannot divide by zero.");
```

```java
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Invalid array index.");

        } catch (Exception e) {
            System.out.println("Error: Invalid input.");

        } finally {
            System.out.println("Finally block executed.");
            sc.close();
        }
    }
}
```

OUTPUT:

```
Enter numerator: 10
Enter denominator: 2
Result = 5
Enter array index: 5
Error: Invalid array index.
Finally block executed.
```